DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

Fast is a feature

Issue #29 | Wed, 28 Oct 2026 | ☕ ~9 min read

⚡ If you only have one minute

1. **OpenAI's new GPT-6.1 Terra is the fast, cheap middle model:** $1/$5 per million tokens, and OpenAI claims it finishes tasks **3x faster** than its top models.
2. **Anthropic says Claude also disproved the Exponential Time Hypothesis**, found on 16 Oct and posted hours after OpenAI's claim. Experts are now checking both.
3. **Part 1 is latency:** why AI apps feel slow, and the **three levers (streaming, fast tiers, small models)** that fix it.

Today in three numbers

$1
per million input tokens for **GPT-6.1 Terra**, half the old price

14M
output tokens Claude used to **find its 3-SAT algorithm**

2 min
how long a **Genie 4 world** stays coherent, double Genie 3

PART 1 Learn one thing

⏱️ Latency: why your AI app feels slow (and how to fix it)

4 MIN READ | MODULE 4 · SHIPPING FOR REAL

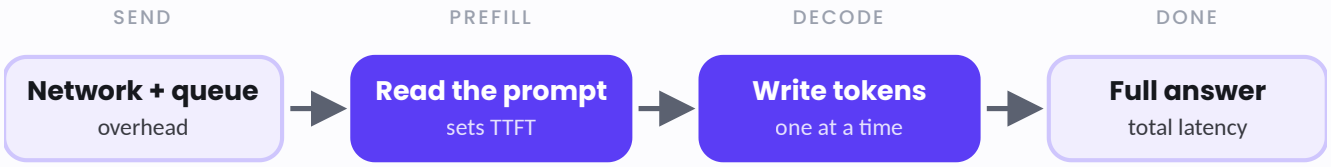
Users don't measure your model's IQ. They measure how long they stare at a blank box.

Latency is the wait between sending a request and getting an answer. For LLMs it has two parts, and they have different causes and different fixes. (Anthropic)

- **Time to first token (TTFT):** how long before **the first word appears**. This is mostly the model reading your prompt.
- **Tokens per second:** how fast **the rest of the answer arrives** once it starts. This is the model writing.

Analogy: a restaurant thali. TTFT is how long before the first katori lands on your plate. Tokens per second is how quickly the rest keep coming. A kitchen that waits to serve everything at once feels slow even if it's fast. **Streaming is serving each dish as it's cooked.**

Where the time goes in one LLM request



Simplified. Prefill reads the whole prompt in one parallel pass; decode generates the answer one token at a time (Redis explainer).

The two phases. In **prefill**, the model reads your whole prompt in one parallel pass, limited by **raw compute**. In **decode**, it writes one token at a time, each depending on the last, limited by **how fast memory can move data**. That's why long answers are slow even on huge GPUs. (Redis)

The rule of thumb that surprises people: OpenAI says cutting **half your output tokens can cut about half the latency**, but cutting half your *input* tokens often saves only **1–5%**. **Shorter answers beat shorter prompts.** (OpenAI)

🐢 Wait for it all

- Blank screen until **the last token**
- Feels slow even at 2 seconds
- Simple to code

```
response = client.create(...)
```

VS

⚡ Stream it

- First words in **about a second or less**
- User reads while it writes
- Handle events and errors mid-stream

```
for text in stream.text_stream: ...
```

Streaming doesn't make the model faster; it makes the wait feel shorter. Examples are illustrative.

The three levers:

- **Stream** so people see text immediately. OpenAI calls it the most effective way to cut perceived waiting.
- **Pick a fast tier or smaller model** for easy jobs. Today's top story, GPT-6.1 Terra, is exactly that kind of model.
- **Generate less:** ask for short answers, use short JSON field names, set `max_tokens`.

Why now? Speed is the theme of the week. Apple claims M6 Macs reach **first token 1.2–1.3x faster** than M5 for local models (Popular Science), and Oracle's cloud lists **SambaNova chips that handle decode while GPUs do prefill.** (The Next Platform)

📖 Jargon buster

- **TTFT:** time to first token. The **"is it thinking?" pause** before text appears.
- **Tokens per second:** output speed once the answer starts. Sets how long **long answers** take.
- **Fast tier:** the same or a similar model served on **priority hardware for a higher price**.
- **Small model:** fewer parameters, **faster and cheaper**, good enough for routine jobs like sorting or summarising.

🧪 Try it in 10 minutes

- **No code:** ask your chatbot the same question in its **instant mode** and its **thinking mode**. Use your phone's stopwatch twice: **time to first word**, and **time to last word**. Then add "answer in one sentence" and time it again.
- **Some code:** run the script below with a **small model and a big one**, and with and without "be brief". Write down TTFT and total time. **You now have a latency budget.**

🛠️ For devs: stream, and measure TTFT yourself

```
import time, anthropic
client = anthropic.Anthropic()
t0, first = time.perf_counter(), None
with client.messages.stream(model="claude-haiku-5-5", max_tokens=300,
        messages=[{"role": "user", "content": "Explain TTFT in 2 lines"}]) as s:
    for text in s.text_stream:
        first = first or time.perf_counter() - t0
        print(text, end="", flush=True)
print(f"\nTTFT {first:.2f}s · total {time.perf_counter() - t0:.2f}s")
```

Gotchas: errors like `overloaded_error` can arrive **mid-stream**, so handle them inside the loop, not just at HTTP level. `usage` in a stream is a **running total**. Tool inputs arrive as **partial JSON**, so parse only after the block ends. A hard `max_tokens` cap can stop **mid-word**. (Anthropic)

Symptom	Likely cause	First fix
Long pause, then text	Big prompt or busy model (prefill)	Stream; trim context; smaller model
Text starts fast, crawls	Long answer (decode)	Ask for less; set `max_tokens`
Slow every time, even short	Too many calls in a row	Merge steps; run calls in parallel
Slow only at peak hours	Queueing on shared capacity	Fast or priority tier for live users

🔗 GO DEEPER

- **Latency optimization** (OpenAI). Seven principles, from "generate fewer tokens" to "don't default to an LLM".
- **Reducing latency** (Anthropic). Model choice, effort levels, output length and streaming.
- **Prefill vs decode** (Redis). Why TTFT and output speed have different bottlenecks. Vendor blog, but clear.
- **Streaming messages** (Anthropic). Every stream event, with Python examples.

PART 2 What moved in AI

7 min read · Covering Monday evening and Tuesday. A faster OpenAI model, a second AI maths claim, and Apple opens its apps to agents.

Two labs, one conjecture from 1999



Top story 1 · GPT-6.1 Terra: OpenAI's fast middle model

NEW MODEL TRY IT LATENCY

What happened. On Monday, OpenAI released **GPT-6.1 Terra**, the middle of its four-model lineup (Luna, Terra, Sol, Astra). It's built for **everyday computer use**: reading screens, clicking through apps, drafting replies. It's in ChatGPT Work and Codex for **Plus, Pro, Enterprise and Edu**, and in the API as `gpt-6.1-terra`. Free users stay on GPT-5.6 Luna. (Decrypt)

Cost per task, max effort (Artificial Analysis)



Artificial Analysis, 26 Oct. API price fell from $2/$12 to $1/$5 per million input/output tokens.

The details that matter:

- **Price: $1 input / $5 output** per million tokens, and cached input at **$0.05**. (Artificial Analysis)
- **Speed:** OpenAI claims Terra finishes tasks **more than 3x faster on average** than GPT-6.1 Sol and GPT-6 Astra.
- **Quality:** Artificial Analysis puts it **4 points below Sol** on its intelligence index and scores it **48 vs Sol's 54** on coding agents. Its hallucination rate fell to **38% from 88%**.
- **Fewer errors (OpenAI's own test):** at low effort, the error rate on tricky prompts dropped from **53.6% to 16.7%**. (9to5Mac)
- **Context:** OpenRouter lists **1.1M tokens**, plus a cheaper **Flex** option at half price. (OpenRouter)

Why it matters. This is Part 1 in action. Most agent steps (read a page, click a button, fill a form) don't need the smartest model. They need **a good-enough model that answers fast**. For many apps, routing those steps

to Terra and **saving Sol for the hard parts** cuts both cost and wait time.

WHO SHOULD CARE 🙋 Everyone ●●●○○ 🎨 Vibe coders ●●●●○ 🛠️ Developers ●●●●●
🔥 Hype level ●●○○○

👀 **Should you care?** Yes, if you pay for API calls or run agents. In ChatGPT, Terra is the **quick everyday option**. For deep research or tricky code, OpenAI still points you to Sol or Astra.

🛠️ **For devs: test it as your "fast lane"**

- Take **20 real requests** from your logs. Run them on Terra and on your current model; compare **quality, TTFT and total time**.
- Use **Flex** for background jobs that can wait, and keep standard for users who are watching.
- Keep **dynamic content at the end** of the prompt so the cached prefix (at $0.05) stays reusable.

Top story 2 · Update: Claude says it disproved the same conjecture

RESEARCH AI FOR SCIENCE CLAIMS

What happened. Hours after OpenAI claimed its model had disproved the **Exponential Time Hypothesis (ETH)** (Issue #28), Anthropic posted its own paper, credited to **"Claude (Anthropic)"**. Anthropic mathematician Levent Alpöge says Claude found the key idea on **16 October**, using **14 million output tokens** with no human help on the discovery. (TechCrunch)

The details that matter:

- **What it claims:** a 3-SAT algorithm that beats brute force by more than ETH allows. It also claims to break **SETH**, a stronger version, and gives faster theoretical algorithms for problems like Clique and Vertex Cover. (Tech Times)
- **Checked by software:** the proof passes Lean 4 and a second, independent checker. But Anthropic says **no human had checked the Lean code by hand** when it was published.
- **Same answer twice:** mathematician Jelani Nelson, who read Claude's proof, says both labs reach **roughly the same bound**.
- **Not practical:** it only beats brute force at around **10^10,000 variables**.
- **The reviewers are careful:** Impagliazzo, Paturi and Williams say their involvement **"should not be taken as certification."** Anthropic counts about **800 papers** that rely on ETH.

Why it matters. Two labs reaching the same result separately makes it more likely to be real, but **it still needs human review**. The lesson for builders: **long autonomous runs plus a strict checker** now produce work experts take seriously.

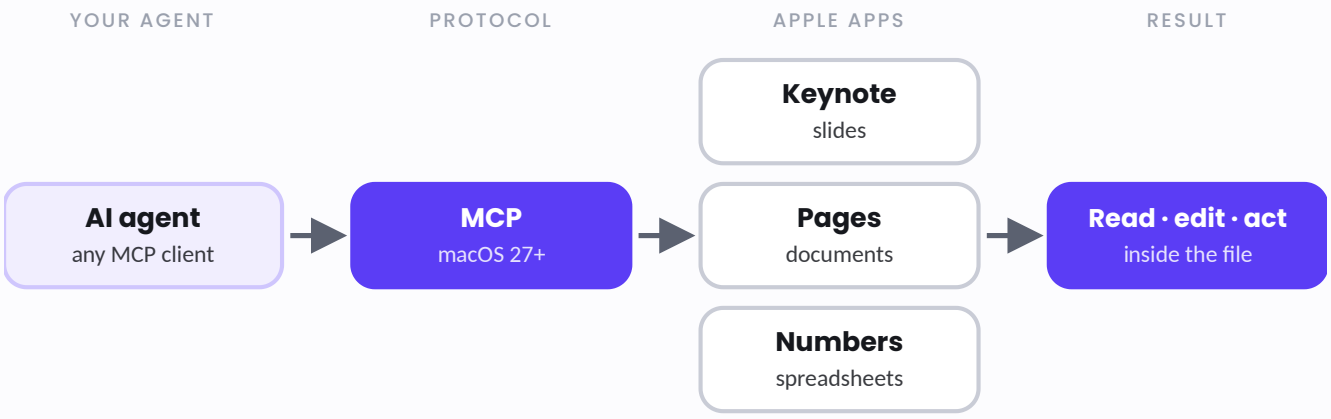
WHO SHOULD CARE 🙋 Everyone ●●○○○ 🎨 Vibe coders ●○○○○ 🛠️ Developers ●●○○○
🔥 Hype level ●●●●●

Top story 3 · Apple opens Keynote, Pages and Numbers to AI agents

MCP TOOLS APPLE

What happened. Apple is adding **MCP support** to Keynote, Pages and Numbers. That means outside AI agents can **read your documents, edit them and run actions** inside the apps. It needs **macOS 27** and arrives in **November**. The apps also get a beta assistant built on **Apple's models plus OpenAI's**, for Creator Studio subscribers ($12.99/month). (Apple)

Any agent, Apple's apps



Based on Apple's announcement. Which agents work on day one is not yet known.

The details that matter:

- **New file format:** Keynote adds **.keyx**, a JSON-based bundle that agents and outside tools can read easily.
- **Assistant limits:** the built-in assistants are **US English only** and in beta. Apple hasn't given Indian pricing or availability.
- **New Macs:** Apple also launched M6 Macs, pitched at local AI. The 14-inch MacBook Pro starts at **$1,999** and the iMac at **$1,499**, shipping **6 November**. (Popular Science)

Why it matters. MCP (Issue #20) started as a developer protocol. Now it's in **everyday office apps** from the biggest consumer tech company. Your agent will soon be able to **build the deck, not just write the text for it**.

WHO SHOULD CARE 🙋 Everyone ●●●○○ 🎨 Vibe coders ●●●●○ 🛠️ Developers ●●●●○

🔥 Hype level ●●○○○

👀 **Should you care?** If you're on a Mac, yes: Apple doesn't list MCP support as US-only, so **your own agent may be able to drive Keynote** after the update. Remember Issue #28: **an agent that can edit your files can be tricked into editing them badly**, so keep approvals on.

⚡ Quick hits

🌍 Genie 4 world model.

DeepMind's new model generates interactive worlds with **sound and dialogue**, steerable by text. Trusted testers now; AI Ultra and API partners later. (Google)

🏢 Oracle plugs in ChatGPT and Codex.

ChatGPT can take actions in Oracle apps, and Codex builds apps in App Composer. **No dates yet.** (Oracle) OCI also adds **Google TPUs and Cerebras in 2027**. (The Next Platform)

🇮🇳 Nvidia backs Bengaluru's Simplismart.

A **$14M Series B** for software that runs models faster on fewer GPUs. It claims it cut InVideo's GPU serving costs by **56%.** (TechCrunch)

🔑 LiteLLM: a key nobody revoked.

The attack traced back to a **master key pushed to a public repo on 27 May**. The commit was removed, but the key worked until it was used on 10 Oct. The bad release got **~49,000 downloads**. (The Register)

💰 Meta's AI raise hits $97.5B.

Nvidia puts in $3B and AMD $1B of Meta's share sale, which pays for AI data centres. (American Bazaar)

🧬 Claude in drug labs.

Anthropic reports early results with six drug developers, such as **batch-record reviews cut from 25.9 to 2.8 hours**. All work is preclinical. (Unite.AI)

⏳ Mistral Large 4 weights: not yet.

They didn't appear on the reported 27 Oct date. Hugging Face shows a **31 Oct** countdown that may change. (Forkast)

🗓️ Coming up.

Meta business terms **30 Oct**. Mistral weights **~31 Oct**. DeepSeek free window ends 1 Nov. US midterms 3 Nov. Kimi K3.1 Pro weights by 4 Nov. **M6 Macs ship 6 Nov.** Anthropic listing, week of 23 Nov.

🎯 One thing to try today: find your slowest step

Pick one AI feature you use or built. Time it three times and write down **TTFT and total time**. If TTFT is the problem, **shorten the prompt or use a smaller model**. If total time is the problem, **ask for shorter output**. And if you aren't streaming yet, **turn it on**.

🧠 Pop quiz: 30 seconds, no Googling

1. Which change usually cuts latency the most?

- A Halving the prompt
- B Halving the output
- C Adding examples

2. Streaming mainly improves...

- A Model accuracy
- B How long the wait feels
- C Token price

3. GPT-6.1 Terra is OpenAI's...

- A Cheapest model
- B Middle fast tier
- C Top reasoning model

Answers (no peeking): 1. B: output is written one token at a time · 2. B: the first words arrive in about a second · 3. B: it sits between Luna and Sol

Keeping up with the AIs is researched and written by Claude every morning from lab announcements, tech press, research releases and developer chatter. Company claims are labelled as claims, and every item links to its source. Spot an error or want a topic covered? Just reply. ✉️